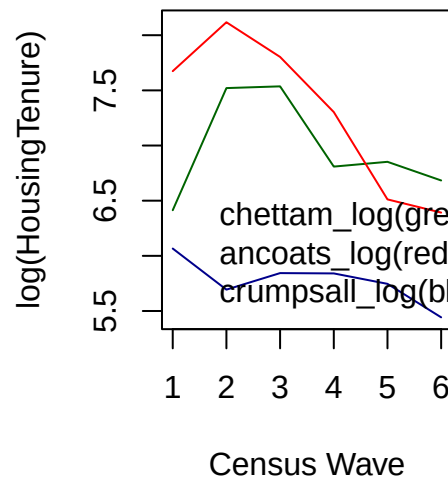
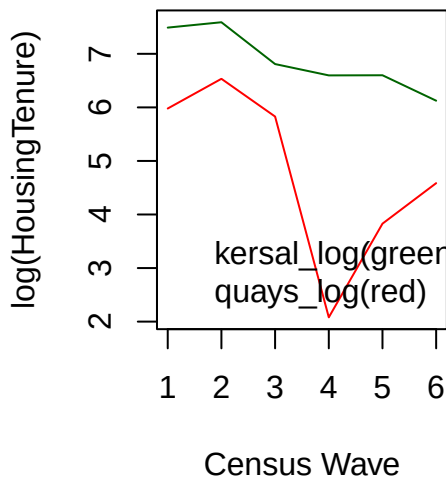


Someplots

```
library(readr)
library(janitor)
library(tidyverse)
library(patchwork)
library(modelsummary)
# Load HousingPanelwide data
HousingPanelwide <- read_csv("d:/PositStuff/gmpanel/HousingPanelwide.csv")
par(mfrow=c(1,2))
ts.plot(cbind(log(HousingPanelwide$kersal.Local Authority etc`),log(HousingPanelwide$`quays
  legend("bottomleft",legend = c("kersal_log(green)","quays_log(red)"),col=c("darkgreen","red")

ts.plot(cbind(log(HousingPanelwide$`chettam.Local Authority etc`),log(HousingPanelwide$`anco
legend("bottomleft",legend = c("chettam_log(green)","ancoats_log(red)","crumpsall_log(blue)"))
```

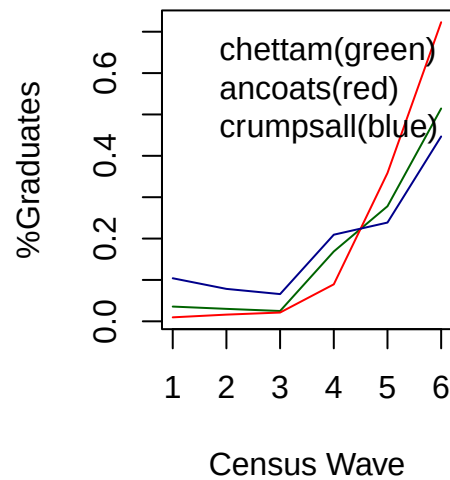
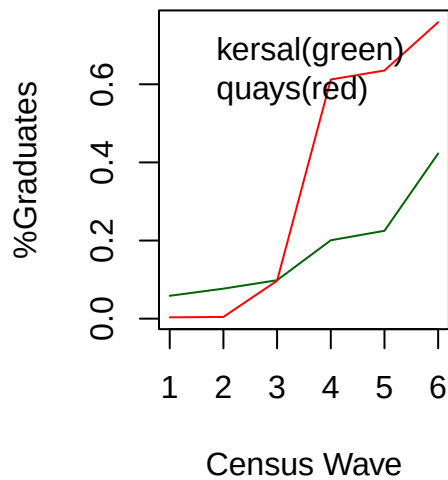
Authority owned tenure 1971–2



```
EducationPanelwide <- read_csv("d:/PositStuff/gmpanel/EducationPanelwide.csv")
par(mfrow=c(1,2))
ts.plot(cbind(EducationPanelwide$`kersal.Graduate`/(EducationPanelwide$kersal.Graduate+EducationPanelwide$`kersal.NonGraduate`),
              EducationPanelwide$`quays.Graduate`/(EducationPanelwide$quays.Graduate+EducationPanelwide$`quays.NonGraduate`)),
        legend("topleft",legend = c("kersal(green)","quays(red)"),col=c("darkgreen","red"),bty = "n")

ts.plot(cbind(EducationPanelwide$`chettam.Graduate`/(EducationPanelwide$chettam.Graduate+EducationPanelwide$`chettam.NonGraduate`),
              EducationPanelwide$`ancoats.Graduate`/(EducationPanelwide$ancoats.Graduate+EducationPanelwide$`ancoats.NonGraduate`),
              EducationPanelwide$`crumpsall.Graduate`/(EducationPanelwide$crumpsall.Graduate+EducationPanelwide$`crumpsall.NonGraduate`)),
        legend("topleft",legend = c("chettam(green)","ancoats(red)","crumpsall(blue)"),col=c("darkgreen","red","blue"),bty = "n")
```

Graduate education 1971–2021 Graduate education 1971–2021 c



```
EducationPanellong <- EducationPanelwide |>
# Create a row identifier
mutate(row_id = row_number()) |>

pivot_longer(
  cols = -c(1, row_id),
  names_to = c("location", "variable"),
  names_pattern = "(.*)\\.(.*)$" #No i do not know how to use pattern matching properly,
) |>

# Pivot wider while respecting the row_id and location mapping
pivot_wider(
  id_cols = c(row_id, location),
  names_from = variable,
  values_from = value
) |>

# 4. Clean up the temporary row_id column
select(-row_id) |>
janitor::clean_names()
```

```
HousingPanellong <- HousingPanelwide |>
# 1. Create a row identifier so each year stays separate
mutate(row_id = row_number()) |>

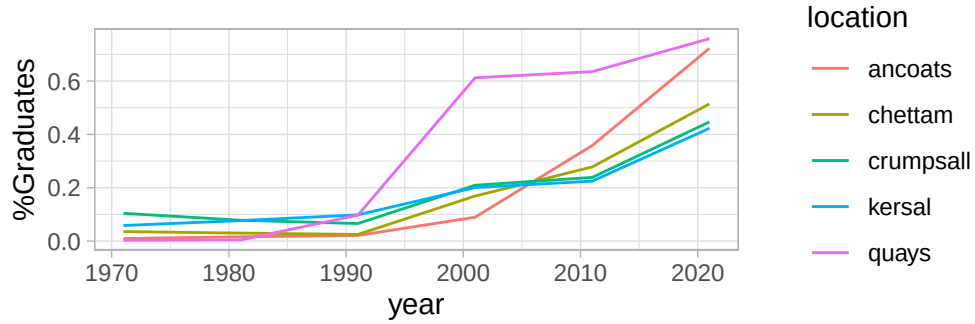
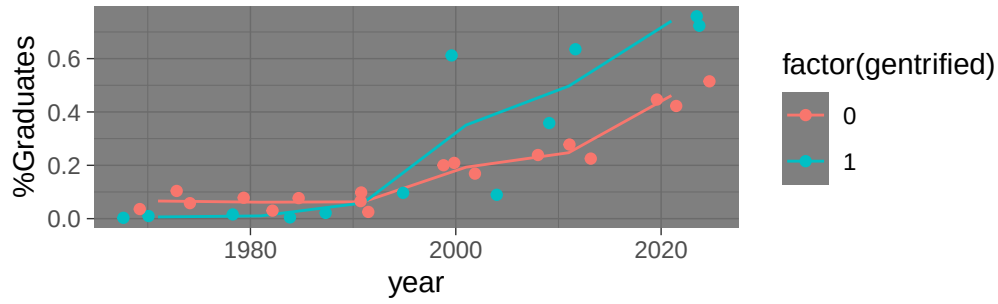
pivot_longer(
  cols = -c(1, row_id),
  names_to = c("location", "variable"),
  names_pattern = "^(.*)\\.\\.(.*)$"
) |>

pivot_wider(
  id_cols = c(row_id, location),
  names_from = variable,
  values_from = value
) |>

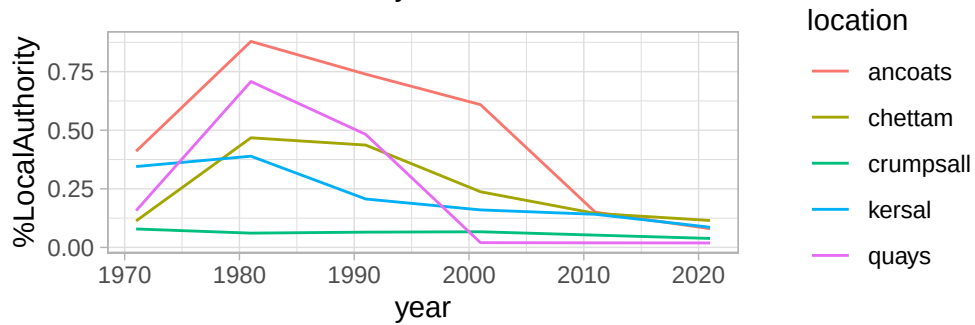
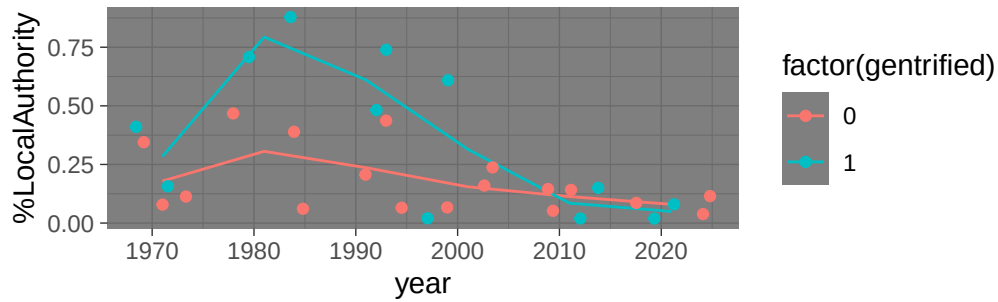
select(-row_id) |>
janitor::clean_names()
```

```
Panellong = merge(HousingPanellong, EducationPanellong) |>
mutate(gentrified=ifelse(location%in%c("ancoats", "quays"),1,0))
```

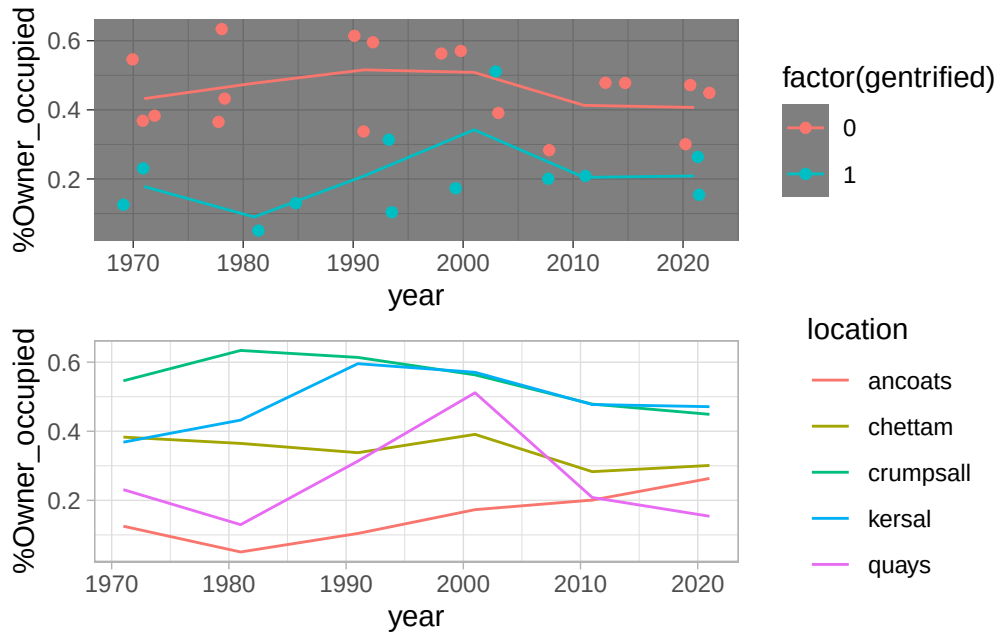
```
p1=ggplot(Panellong,aes(x=year,y=graduate/(graduate+not_graduate),colour=location))+geom_line()
p2=ggplot(Panellong,aes(x=year,y=graduate/(graduate+not_graduate),colour=factor(gentrified)))+geom_line()
p2/p1
```



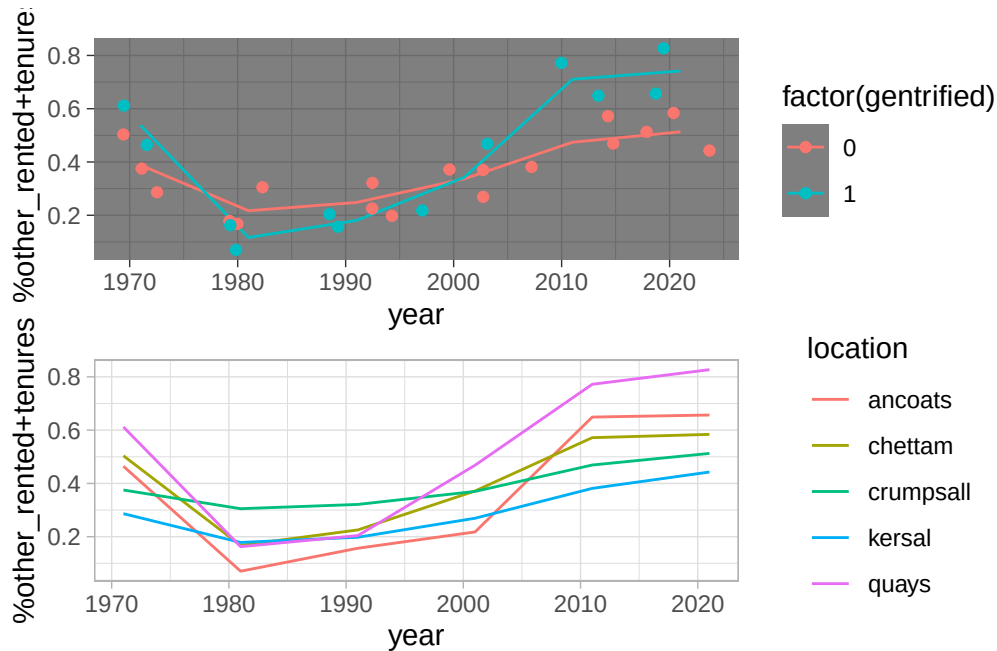
```
p3=ggplot(PanelLong,aes(x=year,y=local_authority_etc/(owner_occupied+other_rented+other_tenur
p4=ggplot(PanelLong,aes(x=year,y=local_authority_etc/(owner_occupied+other_rented+other_tenur
p4/p3
```



```
p5=ggplot(PanelLong,aes(x=year,y=owner_occupied/(owner_occupied+other_rented+other_tenures+1))
p6=ggplot(PanelLong,aes(x=year,y=owner_occupied/(owner_occupied+other_rented+other_tenures+1))
p6/p5
```



```
p7=ggplot(PanelLong,aes(x=year,y=(other_rented+other_tenures)/(owner_occupied+other_rented+other_tenures+1))
  ylab("%other_rented+tenures")+theme_light()
p8=ggplot(PanelLong,aes(x=year,y=(other_rented+other_tenures)/(owner_occupied+other_rented+other_tenures+1))
  stat_summary(fun=mean,geom="line")+geom_jitter()+ylab("%other_rented+tenures")+theme_dark()
p8/p7
```



```
combined_plot = (p4/p3)|(p6/p5)|(p8/p7) + plot_layout(guides = "collect")
modelsummary::datasummary_skim(Panellong)
```

```
EducationPanelProp = EducationPanellong |>
  mutate(GraduateP=graduate/(graduate+not_graduate),
  NGraduateP = not_graduate/(graduate+not_graduate))
HousingPanelProp = HousingPanellong |>
  mutate(local_authority_etcP=local_authority_etc/(owner_occupied+local_authority_etc+other
  owner_occupiedP=owner_occupied/(owner_occupied+local_authority_etc+other_rented+other_ten
  otherrented_tenuresP=(other_rented+other_tenures)/(owner_occupied+local_authority_etc+ot
  LongPanelProp = merge(EducationPanelProp,HousingPanelProp)
  LongPanelPropj= select(LongPanelProp,c(location,year,GraduateP,local_authority_etcP,owner
  mutate(gentrified=ifelse(location%in%c("ancoats","quays"),1,0))
```

```
library(BallMapper)
values = LongPanelPropj$GraduateP
points = LongPanelPropj[,c(4,5,6)]
ball = BallMapper::BallMapper(as.data.frame(points),as.data.frame(values),epsilon = 0.01)
ball = BallMapper::BallMapper(as.data.frame(points),as.data.frame(values),epsilon = 0.17) #e
ballassign = ball$points_covered_by_landmarks

Gentonly = filter(LongPanelPropj,gentrified==1)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
year	6	0	1996.0	17.4	1971.0	1996.0	2021.0	
owner_occupied	30	0	1645.6	1039.8	126.0	1877.0	3247.0	
local_authority_etc	29	0	927.5	817.2	8.0	680.5	3353.0	
other_rented	30	0	1770.1	1251.7	144.0	1612.0	4750.0	
other_tenures	24	0	64.7	61.8	0.0	57.0	206.0	
graduate	30	0	2218.2	2318.8	9.0	1325.0	7804.0	
not_graduate	30	0	14 135.9	22 403.0	254.0	6410.0	82 770.0	
gentrified	2	0	0.4	0.5	0.0	0.0	1.0	
location	N	%						
ancoats	6	20.0						
chettam	6	20.0						
crumpsall	6	20.0						
kersal	6	20.0						
quays	6	20.0						

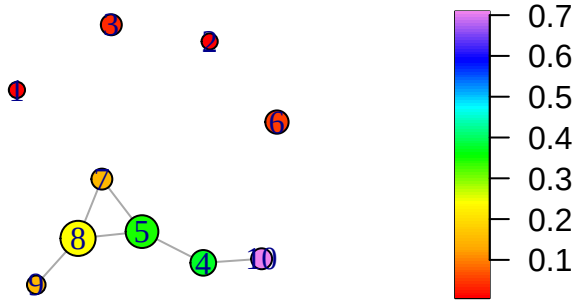
```

values = Gentonly$GraduateP
points = Gentonly[,c(4,5,6)]
ball1 = BallMapper::BallMapper(as.data.frame(points),as.data.frame(values),epsilon = 0.17) #

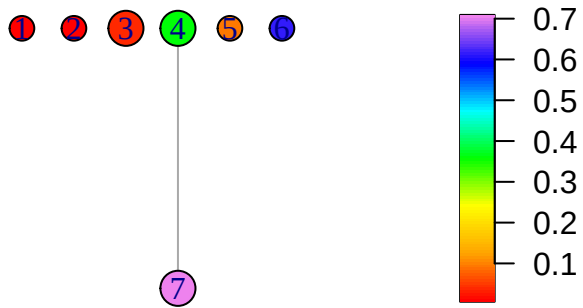
Gentnot = filter(LongPanelPropj,gentrified==0)
values = Gentnot$GraduateP
points = Gentnot[,c(4,5,6)]
ball2 = BallMapper::BallMapper(as.data.frame(points),as.data.frame(values),epsilon = 0.17) #

ColorIgraphPlot(ball)

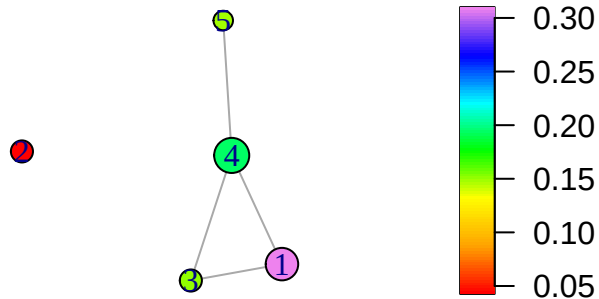
```



```
ColorIgraphPlot(ball1)
```



```
ColorIgraphPlot(ball12)
```



```
#WARNING AI Generated Code as I am only learning how to properly add unit time labels to Bal.
library(igraph)
library(ggraph)
library(ggplot2)

# 1. Number of nodes
n_nodes <- length(ball$coloring)

# 2. Calculate ball sizes from your ballassign indices
ball_sizes <- sapply(ball$points_covered_by_landmarks, length)

# 3. Generate unit-time labels for each ball
node_labels <- sapply(ball$points_covered_by_landmarks, function(indices) {
  subset_data <- PanelLong[indices, ]
  paste0(subset_data$location, "-", subset_data$year, collapse = "\n")
})

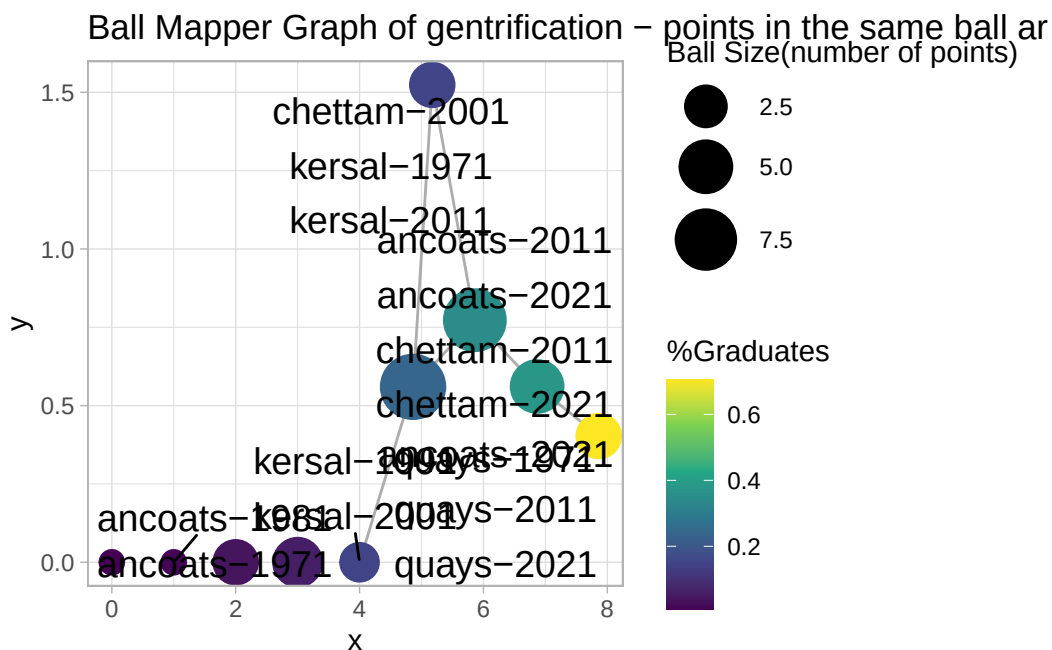
# 4. Build the vertices data frame safely
vertices_df <- data.frame(
  name = 1:n_nodes,
```

```

    label = node_labels,
    coloring = ball$coloring,
    size = ball_sizes
  )

# 5. Create the igraph object
g <- graph_from_data_frame(d = ball$edges, vertices = vertices_df, directed = FALSE)
#Though I did customise the plot below though#
ggraph(g, layout = "stress") +
  geom_edge_link(alpha = 0.8, color = "gray60") +
  geom_node_point(aes(size = size, color = coloring)) +
  geom_node_text(aes(label = label), repel = TRUE, size = 5, color = "black") +
  scale_color_viridis_c() +
  theme_light() +
  labs(title = "Ball Mapper Graph of gentrification - points in the same ball are closest on",
        color = "%Graduates", size = "Ball Size(number of points)") + scale_size_continuous(r

```



```

datasummary_skim(LongPanelPropj)

```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
year	6	0	1996.0	17.4	1971.0	1996.0	2021.0	
GraduateP	30	0	0.2	0.2	0.0	0.1	0.8	
local_authority_etcP	30	0	0.2	0.2	0.0	0.1	0.9	
owner_occupiedP	30	0	0.4	0.2	0.1	0.4	0.6	
otherrented_tenuresP	30	0	0.4	0.2	0.1	0.4	0.8	
gentrified	2	0	0.4	0.5	0.0	0.0	1.0	
location	N	%						
ancoats	6	20.0						
chettam	6	20.0						
crumpsall	6	20.0						
kersal	6	20.0						
quays	6	20.0						